PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The variable trajectory: A systematic review, analytic synthesis and construct domain consolidation of international measures of competence in doctors and medical students.
AUTHORS	Hodgson, Kirsty; Lamport, Daniel; Laville, Allán

VERSION 1 – REVIEW

REVIEWER	Soemantri, Diantha Universitas Indonesia
REVIEW RETURNED	26-Jan-2021

GENERAL COMMENTS	Thank you very much for the opportunity to review this paper. I think the paper is quite well written and the study has been well conducted in terms of the clear and comprehensive methodology and analysis. I do have some thoughts which arises throughout the process of reviewing this paper. Despite the comprehensive and detailed analysis presented by the authors, I wonder whether the paper needs to be strengthened in terms of the introduction as to why such systematic review conducted here is necessary and how the findings of the systematic review can contribute to the literature. I think competence is a very complex nature and consists of many domains, thus it is difficult to say that one measure is better than the other, or is it even necessary to compare measures since the measures are dependent on the competence assessed. Also perhaps an argument needs to be built around the decision of combining both students and doctors measures of competence in the same systematic review. The current trend now is a programmatic, longitudinal assessment, thus relying on one measure of competence would not be suffice. Perhaps how the findings of the systematic review can contribute to that is also worthy to be discussed. In the discussion section: 1. what does it mean by no consensus regarding the definition of competence? Perhaps the authors could elaborate on this, because I think that definition of competence is clear but what comprises competence or what the components are can be varied in some parts from one context to another. 2. it is said that there are likely varying conceptualisations of competence in different culture. I wonder whether it is not only about the culture, but also about the different setting/context where the students are educated/doctors are providing healthcare which will result in perhaps a slightly different list of competence.

	Thank you very much. I hope my comments make sense and can contribute to the improvement of this paper.

REVIEWER	Katowa-Mukwato, Patricia University of Zambia, Basic and Clinical Nursing
REVIEW RETURNED	03-Feb-2021

GENERAL COMMENTS	1. It is expected that medical students should gain competence to a level safe to practice as they graduate, while qualified doctors should remain competent throughout their practice. Therefore, this review is very important as it addressed an important concept "measuring competence in Doctors and Medical students". The review is comprehensive and results are we'll outline in relation to the study objectives. The authors should however address the following; 1.The outcomes of the research are not clearly stated , this is missing throughout the manuscript 2. One study limitations is that English Language restrictions were placed on the search and studies were only from 18 countries. It is important to state specifically where most of the studies reviewed originated, considering that Doctors practice in different and diverse settings as such there may be differences in the contextual definition of competence. Could these contextual differences influence the way competence is measured across the different domains or not? 3. With the complex and diverse nature of medical doctors' competence, did the research team expect to find a single-best-tool for assessing competence? Because from published literature, there is no single best tool for assessing either cognitive or psychomotor competence. Why then should the research team conclude that they did not find a single-best -tool. Such a conclusion may be misleading.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Diantha Soemantri, Universitas Indonesia

Comments to the Author:
Thank you very much for the opportunity to review this paper.

I think the paper is quite well written and the study has been well conducted in terms of the clear and comprehensive methodology and analysis. I do have some thoughts which arises throughout the process of reviewing this paper. Despite the comprehensive and detailed analysis presented by the authors, I wonder whether the paper needs to be strengthened in terms of the introduction as to why such systematic review conducted here is necessary and how the findings of the systematic review can contribute to the literature.
Response: We appreciate the reviewer’s suggestion to clarify the importance of this review and its contribution to the literature.

In order to explore these challenges, and provide an overview of the diverse facets of competence, and their dominance within current assessment measures; the objective of this systematic review of peer-reviewed scientific literature was to identify and evaluate specific measures of competence relevant to doctors and medical students. As the accurate measurement of competence is paramount to patient safety, a systematic review of the measures employed and domains of competence assessed in medical students and doctors was considered important (p.5).

I think competence is a very complex nature and consists of many domains, thus it is difficult to say that one measure is better than the other, or is it even necessary to compare measures since the measures are dependent on the competence assessed. Also perhaps an argument needs to be built around the decision of combining both students and doctors measures of competence in the same systematic review. The current trend now is a programmatic, longitudinal assessment, thus relying on one measure of competence would not be suffice. Perhaps how the findings of the systematic review can contribute to that is also worthy to be discussed.
Response: We agree with the reviewer and have briefly emphasised our rationale on page 5, and added further discussion of this on pages 18 and 21 of the revised manuscript:
Competence is assessed throughout doctors' professional lifespan, from medical student to junior doctor, to senior doctor until retirement (p.5).
However, there has been increased emphasis in more recent studies on assessments which attempt to mimic real world clinical situations (p.18).
There has been increasing interest recently in programmatic longitudinal assessment of both students and doctors using multiple assessment techniques, in which individual assessment elements are part of a process of gathering information contributing to the overall assessment rather than being used on their own to make decisions. In this model these assessments contribute to the learning process, breaking down the traditional dichotomy between formative and summative assessment [13] (p.21).
A wide range of global competence measures were reported in the selected studies. These included written assessments, OSCEs, anatomic simulations, and workplace-based assessments. The combination of a written test with a clinical skill OSCE test has been demonstrated to increase the predictive validity of either alone. In research studies assessing competence there is considerable interest in simulation-based assessments and some have focussed on cognitive assessment which may be of particular importance with many doctors working beyond retirement (p.21).

In the discussion section:
1. what does it mean by no consensus regarding the definition of competence? Perhaps the authors could elaborate on this, because I think that definition of competence is clear but what comprises competence or what the components are can be varied in some parts from one context to another.
Response: We appreciate the reviewer raising this important point, we too were surprised that in the selected literature we found no consensus on an explicit and standardised definition of competence. However, we agree that the components relevant to conceptualisation are pertinent in this review. We have therefore replaced the word definition with conceptualisation to reflect this:
Therefore, although competence is considered paramount for safe practice, no consensus in the selected literature exists regarding its conceptualisation, nor is there measurement standardisation (p.20).

2. it is said that there are likely varying conceptualisations of competence in different culture. I wonder whether it is not only about the culture, but also about the different setting/context where the students are educated/doctors are providing healthcare which will result in perhaps a slightly different list of competence.
Response: We thank the reviewer, and agree that this is an important point. We have amended the text to this effect on page 20:

There are likely varying conceptualisations of competence in different cultures, and in different clinical, social and economic contexts, therefore further research on geographic and cultural construct analysis is advocated to explore potential differences in worldwide assessments.

Thank you very much. I hope my comments make sense and can contribute to the improvement of this paper.

Reviewer 2

Reviewer: 2
Dr. Patricia Katowa-Mukwato, University of Zambia

Comments to the Author:

1. It is expected that medical students should gain competence to a level safe to practice as they graduate, while qualified doctors should remain competent throughout their practice. Therefore, this review is very important as it addressed an important concept "measuring competence in Doctors and Medical students". The review is comprehensive and results are we'll outline in relation to the study objectives. The authors should however address the following;

1.The outcomes of the research are not clearly stated , this is missing throughout the manuscript
Response: We thank the reviewer for the opportunity to clarify the objectives and outcomes of this systematic review of the literature. We have now summarised the main outcomes for clarity throughout the manuscript:
Therefore, the outcome of the construct domain consolidation of measures of competence was the identification of 12 domains. There were significant differences in the predominant domains assessed in IM versus RM studies, with less focus on professionalism and knowledge-based domains in the latter (p.17).
Therefore, many assessments of competence are contingent upon assessor subjectivity. Whilst some studies demonstrated attempts to assess inter-rater reliability, this is an important issue that requires further attention (p.20).

2. One study limitations is that English Language restrictions were placed on the search and studies were only from 18 countries. It is important to state specifically where most of the studies reviewed originated, considering that Doctors practice in different and diverse settings as such there may be differences in the contextual definition of competence. Could these contextual differences influence the way competence is measured across the different domains or not?
Response: We acknowledge that the inclusion of publications only in the English language or with English language translations in this systematic review is a limitation. We appreciate the reviewer highlighting this important point, which is now emphasised on page 9 of the revised manuscript with the inclusion of the geographical origin of the publications:
Included articles were published between 1990 and 2020. There was a progressive increase in the number of publications per year over this period (mean annual publications per year 0.3 from 1990 to 1999, 1.9 from 2000 to 2009, and 8.2 from 2010 to 2019), consistent with increasing research interest and societal awareness of the extent and consequences of medical error[P105]. The study designs included: two intervention, 22 measurement development, 26 measurement evaluation, 17 curriculum

development, 40 clinical assessment, 21 training evaluation, and 11 experiential assessment; 26 studies comprised multiple categories.

Studies were restricted to English language, but without restriction on geographic origin and were derived from 18 countries. One study[P26] undertook data collection in two countries (USA and Canada), therefore data was counted in both. The majority of research was from the USA (n = 34 articles), UK (n = 17), Germany (n = 15), and Canada (n = 15). There were five studies from Australia[P3,35,64,80,102], four from the Netherlands[P23,49,58,69], and three from Korea[P13,61,73]. Two studies were from New Zealand[P5,12] and Denmark[P45,66]. Sweeden[P10], Lebanon[P36], Saudi Arabia[P46], Austria[P48], Belgium[P50], Norway[P59], Poland[P60], India[P79], and China[P103] were each represented by a single study. The percentage distribution of all articles geographic origin is presented in Figure 2.

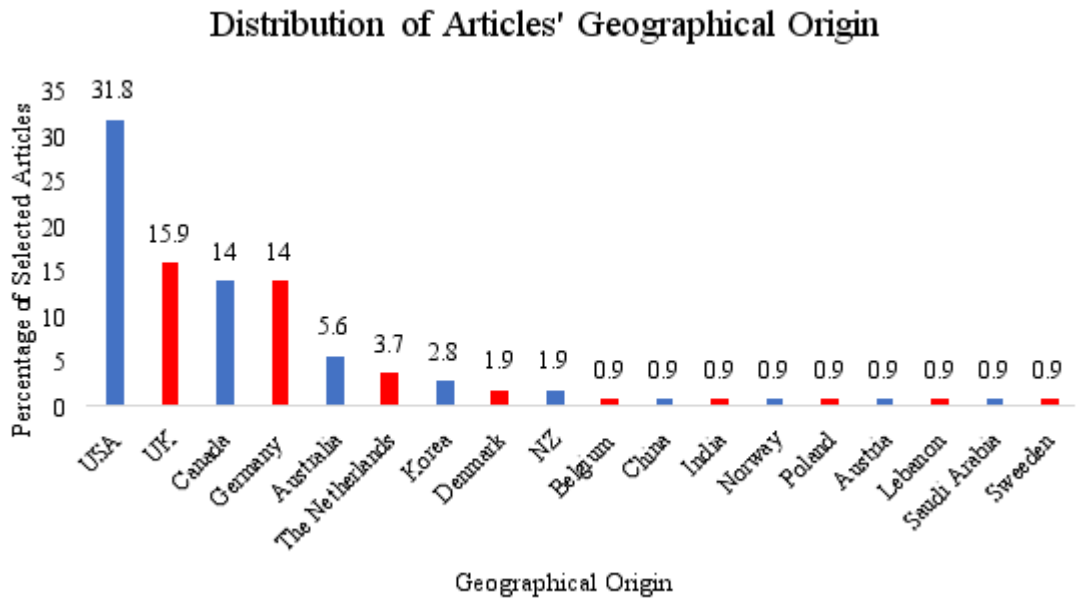


Figure 2 The Percentage Distribution of Selected Articles' Geographical Origin

3. With the complex and diverse nature of medical doctors' competence, did the research team expect to find a single-best-tool for assessing competence? Because from published literature, there is no single best tool for assessing either cognitive or psychomotor competence. Why then should the research team conclude that they did not find a single-best -tool. Such a conclusion may be misleading.

Response: We appreciate the reviewer raising this point and agree that it is not surprising given the complex interacting components (both trait and state) that determine competence, however traditional assessments of competence have not always recognised this. The complexity and diversity inherent in a doctor's career make this a complex task. This sentence has been removed as recommended.

VERSION 2 – REVIEW

REVIEWER	Soemantri, Diantha Universitas Indonesia
REVIEW RETURNED	30-Apr-2021

GENERAL COMMENTS	Thank you to the authors for the revision. I think the paper has significantly improved. I still have some minor suggestions though, first is regarding the inclusion of papers on selection of medical

	student candidates. I am not really sure that it matches the objective of this study. Line 3-7 on page 20: shouldn't it be in the discussion? Line 23-31 on page 21: the highlighted sentences seem to be a little bit detached from the paragraph? Perhaps more explanation on how the findings of this review can contribute to the development and implementation of programmatic assessment. Thank you very much, hope my comments help.

VERSION 2 – AUTHOR RESPONSE

Round 2 Comments:

31-Jul-2021

Reviewer: 1
Dr. Diantha Soemantri, Universitas Indonesia

Comments to the Author:

Thank you to the authors for the revision. I think the paper has significantly improved. I still have some minor suggestions though, first is regarding the inclusion of papers on selection of medical student candidates. I am not really sure that it matches the objective of this study.

Response: We appreciate the reviewer raising this important point. We believed it was appropriate to include assessment of competence in medical student candidates in this systematic review in order to identify baseline assessments and examine the domains relevant to competence assessment accross the professional lifespan. Whilst the assessments must be tailored to both the stage and in some cases the specialty of the doctor, the process is therefore a continuum. Assessment of competence is crucial for patient safety and is undertaken throughout a doctor's medical career. Whilst many of the skills required of a doctor can be taught, it is clear that there are substantial individual variations in attributes such as coping that underpin competence. Medical schools seek to assess some of these as part of student selection. We therefore felt it important to include assessment of these aspects of competence in student selection, since this forms the beginning of this continuous process of comptence. Of note, however, is the dirth of studies that have rigorously assessed the validity of these measures (three identified in this systematic review), suggesting that this is an area in need of further research and validation of the predictive value of these assessments in determining competence throughout a doctors career. The manuscript has been revised to clarify this rationale.

Competence is assessed throughout doctors' professional lifespan, from selection for entry to medical school to medical student to junior doctor, to senior doctor until retirement. (p.5)

As the accurate measurement of competence is paramount to patient safety, a systematic review of the measures employed and domains of competence assessed in medical student selection, medical students and doctors was considered important. (p.5)

The assessment of some domains such as coping competence are considered by many medical schools to be crucial in identifying those with the necessary attributes to train to be a doctor. These domains may therefore be considered as part of the continuum of assessment of competence throughout a medical career. These were therefore included in this systematic review. This continuum approach is consistent with the increasing recent interest in programmatic longitudinal assessment of both students and doctors using multiple assessment techniques, in which individual assessment

elements are part of a process of gathering information contributing to the overall assessment rather than being used on their own to make decisions. (p.21).

Line 3-7 on page 20: shouldn't it be in the discussion?

Response: We thank the reviewer for their recommendation, this has now been revised and moved to page 21 of the discussion.

Many assessments of competence are contingent upon assessor subjectivity. Whilst some studies demonstrated attempts to assess inter-rater reliability, this is an important issue that requires further attention. Assessor bias is a worthwhile area of further research due to concerning discrepancies in assessment accuracy that have been attributed to unreliable self-assessment,[14] inconsistent applications of organizational standards resulting in a lack of reliability, and unclear rating scale criteria. (p.21)

Line 23-31 on page 21: the highlighted sentences seem to be a little bit detached from the paragraph? Perhaps more explanation on how the findings of this review can contribute to the development and implementation of programmatic assessment.

Response: We appreciate the reviewer's suggestion, and have repositioned and contextualised this section.

The assessment of some domains such as coping competence are considered by many medical schools to be crucial in identifying those with the necessary attributes to train to be a doctor. These domains may therefore be considered as part of the continuum of assessment of competence throughout a medical career. These were therefore included in this systematic review. This continuum approach is consistent with the increasing recent interest in programmatic longitudinal assessment of both students and doctors using multiple assessment techniques, in which individual assessment elements are part of a process of gathering information contributing to the overall assessment rather than being used on their own to make decisions. In this model these assessments contribute to the learning process, breaking down the traditional dichotomy between formative and summative assessment [13]. The findings of this systematic review support this, because it is clear that individual assessments taken in isolation each have their limitations. Furthermore, the inclusion of both formative and summative components in this longitudinal assessment enhances the learning process. (p.21)

Thank you very much, hope my comments help.

Response: We thank the reviewer for their valuable comments and suggestions.